

Explainable Deep Learning Framework for Pneumonia Detection and Classification Using Transfer Learning and Grad-CAM

Abstract—Pneumonia remains a major respiratory disease requiring timely and accurate diagnosis, particularly in clinical settings where radiological expertise and diagnostic resources may be limited. Chest X-ray imaging is widely used as a first-line diagnostic modality due to its accessibility, low cost, and rapid acquisition. However, reliable interpretation of chest X-ray images is challenging because visual manifestations of COVID-19, bacterial pneumonia, viral pneumonia, and normal lung conditions may overlap. Recent advances in deep learning have demonstrated strong potential for automated chest disease detection, but many existing studies focus mainly on binary classification and provide limited clinical interpretability. This paper proposes an explainable deep learning framework for four-class chest X-ray classification involving Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia cases. The proposed study evaluates three modern convolutional architectures: DenseNet121, EfficientNetV2B3, and EfficientNetV2S. Transfer learning is employed to improve model convergence and performance under medical image data constraints. Grad-CAM is integrated to provide visual explanations by highlighting image regions that influence model predictions. In addition to classification, the proposed system includes web-based and mobile-friendly deployment, historical prediction management, and automated PDF medical report generation. Experimental results show that DenseNet121 achieved the highest accuracy of 85.31%, followed by EfficientNetV2B3 with 84.30% and EfficientNetV2S with 80.54%. The findings demonstrate that explainable deep learning can support clinical decision-making by combining predictive performance, transparency, and practical deployment features. The proposed framework contributes toward an end-to-end clinical decision support system for pneumonia detection and classification using chest X-ray images.

Keywords—Clinical Decision Support, COVID-19, Deep Learning, Grad-CAM, Pneumonia Detection

I. INTRODUCTION

Pneumonia is one of the leading causes of morbidity and mortality worldwide, responsible for millions of deaths annually and disproportionately affecting children under five years of age and elderly populations [1]. It is caused by a variety of pathogens including bacteria, viruses, fungi, and atypical organisms, each presenting with distinct radiographic and clinical characteristics that require different treatment approaches. The timely and accurate differentiation of pneumonia subtypes including bacterial pneumonia, viral pneumonia, and COVID-19 from normal lung conditions is therefore a clinical necessity that directly influences patient management, antibiotic prescription decisions, and isolation protocols. Chest X-ray imaging remains the most widely adopted first-line diagnostic modality for pulmonary diseases due to its accessibility, low cost, rapid acquisition, and broad availability across clinical settings, including low-resource environments. However, reliable interpretation of chest X-ray images presents substantial challenges even for experienced radiologists. Visual manifestations of COVID-19, bacterial pneumonia, viral pneumonia, and normal lung conditions

frequently overlap, and radiographic findings can be subtle or non-specific, particularly in early-stage infections. Radiologist availability is uneven across institutions and geographic regions, and high patient volumes in emergency and intensive care settings can introduce diagnostic delays that adversely impact patient outcomes. Automated and explainable computational tools therefore offer critical support for reducing radiologist workload, improving diagnostic consistency, and providing rapid preliminary assessments in high-volume clinical environments.

Deep learning has emerged as the dominant paradigm in medical image analysis, as convolutional neural networks (CNNs) automatically extract hierarchical visual representations from low-level edges and textures to high-level structural patterns without the need for handcrafted feature engineering [2]. Foundational research established that deep networks achieve strong image recognition performance given appropriate architectures and training strategies [2], motivating the development of AI systems capable of supporting clinical authority while reducing diagnostic burden [1], [3]. Large benchmark datasets have greatly accelerated chest X-ray research. The ChestX-ray8 dataset established a hospital-scale multi-label thoracic abnormality benchmark comprising over 100,000 images across 14 disease labels [4], and CheXpert extended this paradigm with uncertainty labels and expert comparison protocols [5]. Dedicated COVID-19 chest X-ray datasets further enabled the rapid development of automated screening models during the pandemic [6]. Nevertheless, most existing studies remain confined to binary or limited-class classification rather than a clinically meaningful multi-class distinction encompassing Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia as distinct diagnostic categories.

Prior pneumonia detection studies have demonstrated encouraging results: CheXNet achieved radiologist-level pneumonia detection using DenseNet-based learning [7], and numerous COVID-19 studies applied transfer learning for binary or limited classification tasks [8], [9], [10]. However, direct comparison across studies is difficult due to heterogeneous datasets, preprocessing pipelines, and evaluation protocols. Furthermore, black-box model predictions without interpretability remain a significant barrier to clinical adoption, as clinicians require evidence that model attention is grounded in genuine pathological findings rather than spurious visual artifacts unrelated to the disease process. Transfer learning is especially valuable in medical imaging, where acquiring large annotated datasets is costly and time-consuming. Pretraining on large natural image datasets enables effective feature adaptation through selective fine-tuning [11], [12]. Explainability is equally essential for clinical trust, as Grad-CAM generates class-discriminative heatmaps identifying image regions most influential to model predictions [13]. Medical XAI research confirms that interpretability enables clinicians to assess whether models focus on clinically relevant anatomical regions [14], and

broader XAI frameworks emphasize transparency in high-stakes patient-care domains [15].

This study addresses five key gaps: the predominance of binary classification; limited comparative evaluation of DenseNet121, EfficientNetV2B3, and EfficientNetV2S on a unified dataset and training pipeline; insufficient Grad-CAM integration as a workflow-embedded feature; and the absence of fully deployable clinical decision support systems incorporating secure access, historical prediction management, and automated reporting. The proposed framework performs four-class classification (Normal, COVID-19, Bacterial Pneumonia, Viral Pneumonia), comparatively evaluates the three architectures with transfer learning, integrates Grad-CAM explainability, and delivers an end-to-end web- and mobile-compatible platform with role-based access, prediction history, and automated PDF report generation.

II. LITERATURE REVIEW

Deep learning in computer vision has advanced through a series of landmark architectural developments. The introduction of residual learning resolved the degradation problem in very deep networks through skip connections, enabling stable training of networks exceeding 100 layers and achieving substantial improvements over prior architectures [16]. Prior milestones including deep CNNs with nonlinear activations and dropout regularization, multi-scale inception modules for efficient feature extraction, and very deep networks with small convolutional filters collectively established the representational capacity and training paradigms that underpin modern medical image classification. Visualization techniques further clarified how intermediate convolutional layers respond selectively to image patterns, providing early insight into the learned feature hierarchy of deep networks [17].

DenseNet introduced dense connectivity in which each layer receives feature maps from all preceding layers within a dense block, promoting feature reuse, improved gradient flow throughout the network, and reduced parameter redundancy compared to conventional architectures [18]. This architectural property is particularly advantageous for chest X-ray analysis, where multi-scale feature information is necessary to differentiate between subtle disease-specific patterns such as opacity distributions, peribronchial thickening, airspace consolidation, and ground-glass opacities characteristic of different pneumonia subtypes. DenseNet121 has proven effective in medical imaging tasks due to its capacity for fine-grained pattern capture and parameter efficiency, as evidenced by the influential CheXNet study that demonstrated densely connected networks achieving radiologist-level performance in pneumonia detection [7].

EfficientNet proposed a principled compound scaling methodology that simultaneously and uniformly scales network depth, width, and input resolution using a fixed set of scaling coefficients [19], overcoming the limitations of ad hoc scaling strategies that independently adjust single dimensions at the expense of others. EfficientNetV2 subsequently refined the original EfficientNet family by introducing Fused-MBConv blocks in earlier network stages, progressive learning with adaptive regularization schedules, and substantially improved training efficiency without sacrificing accuracy [20]. EfficientNetV2B3 is well-suited to deployment-oriented medical imaging applications due to its

strong accuracy-to-computational-cost ratio. Prior work on ensemble strategies combining EfficientNetV2-based architectures has demonstrated competitive performance in multiclass chest disease classification, including COVID-19 and other pulmonary conditions [21], supporting the inclusion of EfficientNetV2 variants in this comparative study.

ImageNet-based pretraining [22] has become the standard initialization strategy in medical image analysis, supported by both theoretical justification and empirical evidence. Pan and Yang formally defined transfer learning as leveraging knowledge from a source domain to improve learning performance in a related target domain [11], establishing the conceptual foundation for applying natural image pretrained models to medical tasks. Tajbakhsh et al. empirically demonstrated that fine-tuning pretrained CNNs is consistently more effective than training from scratch across a wide range of medical image analysis tasks [12]. Although ImageNet contains natural rather than medical images, early convolutional layers learn generic visual features including edges, textures, and spatial gradients that are transferable across domains. Comprehensive surveys confirm that CNNs have been successfully applied to classification, detection, segmentation, and clinical prognosis in medical imaging, while emphasizing the need for careful methodology to address dataset biases, class imbalance, and spurious correlations [23].

Large chest X-ray benchmarks such as ChestX-ray8 [4] and CheXpert [5] primarily support multi-label thoracic abnormality recognition rather than focused pneumonia subtype classification. Prior studies including CheXNet [7], Apostolopoulos and Mpesiana [8], Ozturk et al. [9], and Narin et al. [10] demonstrated the feasibility of deep learning for pneumonia detection but did not fully address the clinically important four-class distinction among Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia within a unified framework, directly motivating the proposed study. Explainability is a core requirement in medical deep learning. Class activation mapping introduced discriminative region localization in CNNs [17], and Grad-CAM extended this using final convolutional layer gradients to produce precise class-discriminative heatmaps without architectural modifications [13]. Medical XAI surveys stress that interpretability is essential for clinical acceptance and responsible deployment [14], while broader XAI frameworks affirm its indispensability in high-stakes decision-making [15]. Topol argued that AI should augment rather than replace clinicians [1], and Shortliffe and Sepúlveda emphasized the need for interpretable, workflow-integrated outputs for meaningful clinical adoption [3].

Based on the reviewed literature, five key gaps are identified: (i) most existing studies are binary and do not address multiclass pneumonia subtype classification; (ii) the four-class distinction among Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia remains underexplored in a unified framework; (iii) few studies compare DenseNet121, EfficientNetV2B3, and EfficientNetV2S under the same dataset and training pipeline; (iv) explainability is typically presented as an add-on visualization rather than a workflow-integrated feature with patient record linkage; and (v) deployment aspects including secure authentication, role-based dashboards, historical prediction management, and automated report generation are frequently absent from

published systems. The proposed framework directly and systematically addresses all five identified gaps.

III. METHODOLOGY

The proposed methodology encompasses five principal stages: dataset preparation and analysis, preprocessing and augmentation, transfer learning-based model training, comparative evaluation with Grad-CAM explainability integration, and end-to-end clinical system deployment. The overarching objective is to classify chest X-ray images into four clinically meaningful categories Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia while ensuring fair and reproducible comparison among DenseNet121, EfficientNetV2B3, and EfficientNetV2S. A standardized training and evaluation framework is adopted across all architectures to eliminate confounding variables introduced by inconsistent experimental conditions and to ensure that observed performance differences are attributable to architectural properties rather than training artifacts.

A. Dataset Description

The dataset used in this study comprises a total of 9,208 chest X-ray images drawn from publicly available medical imaging repositories [4], [6], [7]. The images span four diagnostic categories: COVID-19, Normal, Viral Pneumonia, and Bacterial Pneumonia, as detailed in Table I. The dataset exhibits notable class imbalance, with the Normal and Bacterial Pneumonia classes substantially over-represented relative to the COVID-19 and Viral Pneumonia classes. This imbalance reflects natural prevalence distributions in clinical settings but necessitates explicit mitigation strategies, including class-weighted loss computation and stratified sampling, to prevent the classifier from developing a bias toward majority classes. All images are in JPEG format and vary in acquisition resolution, contrast, and pixel intensity characteristics arising from differences in imaging equipment, patient demographics, and clinical acquisition conditions, further motivating systematic preprocessing and normalization procedures. Table I summarizes the class-wise distribution of the 9,208 chest X-ray images used in this study.

TABLE I. NUMBER OF IMAGES PER CLASS IN THE DATASET

Class	Number of Images
COVID-19	1,281
Normal	3,270
Viral Pneumonia	1,656
Bacterial Pneumonia	3,001
Total	9,208

B. Data Preprocessing

Chest X-ray images vary substantially in resolution, contrast, and pixel intensity due to differences in imaging equipment and acquisition protocols, necessitating a systematic preprocessing pipeline applied uniformly across all subsets. Images are resized to the input resolution required by each architecture, with DenseNet121 receiving 224×224 pixel inputs and EfficientNetV2B3 and EfficientNetV2S configured to their respective optimal resolutions, ensuring fair comparison through consistent data partitions and architecture-specific normalization. Since chest X-ray images are inherently grayscale while ImageNet-pretrained models expect three-channel RGB inputs, grayscale images are converted through channel replication, repeating the single luminance channel across all three color channels to enable reuse of pretrained weights without architectural

modification. Pixel values are normalized using channel-wise mean subtraction and standard deviation scaling consistent with ImageNet training distribution statistics. A data quality audit is performed prior to training to remove duplicate images, corrupted files, and samples below a minimum resolution threshold, preventing data leakage and ensuring evaluation metrics reflect clean, representative conditions.

C. Data Splitting and Augmentation

The dataset is partitioned into training, validation, and testing subsets using stratified random sampling to ensure class distribution is preserved across all subsets, preventing biased performance estimates under class imbalance. The training set supports model weight optimization, the validation set guides hyperparameter selection and early stopping, and the held-out test set is reserved exclusively for final unbiased evaluation. Data augmentation is applied exclusively to the training set and includes small random rotations, horizontal and vertical translations, zoom augmentation, brightness adjustment, and mild contrast variation to simulate realistic acquisition variability and improve model generalization. Horizontal flipping is applied cautiously given chest X-ray anatomical laterality, and excessive augmentation that could distort diagnostically meaningful structures is deliberately avoided. Validation and test images undergo only deterministic resizing and normalization, ensuring evaluation metrics reflect performance on unmodified images representative of real clinical inputs.

D. Model Architectures

Three architectures representing distinct design philosophies are selected for comparative evaluation. DenseNet121 implements dense connectivity in which each layer is directly connected to every subsequent layer within a dense block [18], promoting multi-level feature reuse, reducing the vanishing gradient problem, and retaining both fine-grained and abstract representations throughout the network. These properties are particularly beneficial for chest X-ray classification, where diagnostic information spans multiple spatial scales. Its final classification layer is replaced with a four-neuron softmax output while the pretrained backbone is preserved.

EfficientNetV2B3 is selected for its compound scaling strategy and superior training efficiency [20], introducing Fused-MBConv blocks and progressive learning to improve training throughput and reduce overfitting. Its balanced depth, width, and resolution configuration suits web and mobile clinical deployment, and its backbone is adapted with a custom head comprising global average pooling, dropout, a dense intermediate layer, and a four-class softmax output. EfficientNetV2S, the larger variant within the same family [20], is included to evaluate whether its increased representational capacity and greater proportion of Fused-MBConv blocks provide meaningful accuracy gains over EfficientNetV2B3, enabling a within-family comparison that complements the cross-family analysis. Its pretrained classifier is similarly replaced with an identical four-class softmax head.

E. Transfer Learning Strategy

Transfer learning proceeds in two sequential stages designed to balance feature adaptation with regularization. In the first stage, all pretrained backbone layers are frozen and only the custom classification head is trained. This allows the

randomly initialized head to converge toward task-specific decision boundaries while the high-quality pretrained feature representations remain intact, preventing large gradient updates from destabilizing pretrained weights before the classification head reaches a stable solution. In the second stage, selected deeper backbone layers are progressively unfrozen and fine-tuned at a substantially reduced learning rate, allowing high-level feature representations to gradually adapt to chest X-ray characteristics while lower-level generic features are largely preserved. This two-stage strategy is justified by established transfer learning theory [11] and empirical evidence from medical image analysis [12]: ImageNet-pretrained CNNs [22] capture generalizable visual representations that provide a strong initialization, and careful fine-tuning progressively specializes these representations toward the medical imaging domain without catastrophic forgetting of pretrained knowledge. The same two-stage transfer learning protocol is applied consistently across all three architectures to enable fair comparative evaluation.

F. Hyperparameter Configuration

All models are trained using categorical cross-entropy loss for four-class classification and optimized using the Adam optimizer. Training is performed in two stages: an initial classifier-head training phase with a learning rate of 1×10^{-4} , followed by fine-tuning with a reduced learning rate of 1×10^{-5} . Dropout layers are incorporated in the classification head to mitigate overfitting, and a batch size of 32 is used throughout. Early stopping is applied based on validation loss, while a ReduceLROnPlateau scheduler automatically decreases the learning rate when validation improvement stagnates over a defined patience window. Class weights are computed from the training set distribution and applied during loss computation to address class imbalance and prevent majority-class bias. Model checkpoints are saved based on best validation accuracy. To ensure fair comparison, the same train-validation-test split, augmentation strategy, optimization settings, stopping criteria, and evaluation metrics are applied uniformly across all three architectures.

G. Evaluation Metrics

Performance is evaluated using accuracy, precision, recall, F1-score, and confusion matrix analysis. Overall accuracy provides a global correctness measure but is insufficient under class imbalance; per-class precision assesses the reliability of positive predictions for each class, per-class recall measures sensitivity to actual disease cases, and per-class F1-score balances both through their harmonic mean. The confusion matrix reveals which diagnostic categories are most frequently confused, providing qualitative insight into classification errors. Class-wise recall is particularly critical in medical imaging, because missed pneumonia or COVID-19 cases can lead to delayed treatment with serious clinical consequences. Final model selection therefore balances high overall accuracy, stable class-wise performance, explainability quality, and practical deployment feasibility.

H. Grad-CAM Explainability

Grad-CAM is integrated to generate visual explanations by computing gradients of the predicted class score with respect to the final convolutional feature maps for a given input image. These gradients are globally average-pooled to produce layer-specific importance weights, which are linearly combined with the corresponding activation maps and passed through a ReLU operation to retain only positively contributing features. The resulting heatmap is upsampled to the original image resolution and overlaid on the chest X-ray image using a color scale to indicate regions most influential to the model's prediction. Clinically meaningful Grad-CAM maps are expected to highlight relevant lung parenchyma regions rather than irrelevant borders, image labels, or extra-thoracic structures. Grad-CAM is not intended to replace expert radiological interpretation but to improve prediction transparency and serve as visual decision-support evidence. Each system prediction includes the class label and corresponding Grad-CAM visualization to support clinical review.

I. System Deployment Architecture

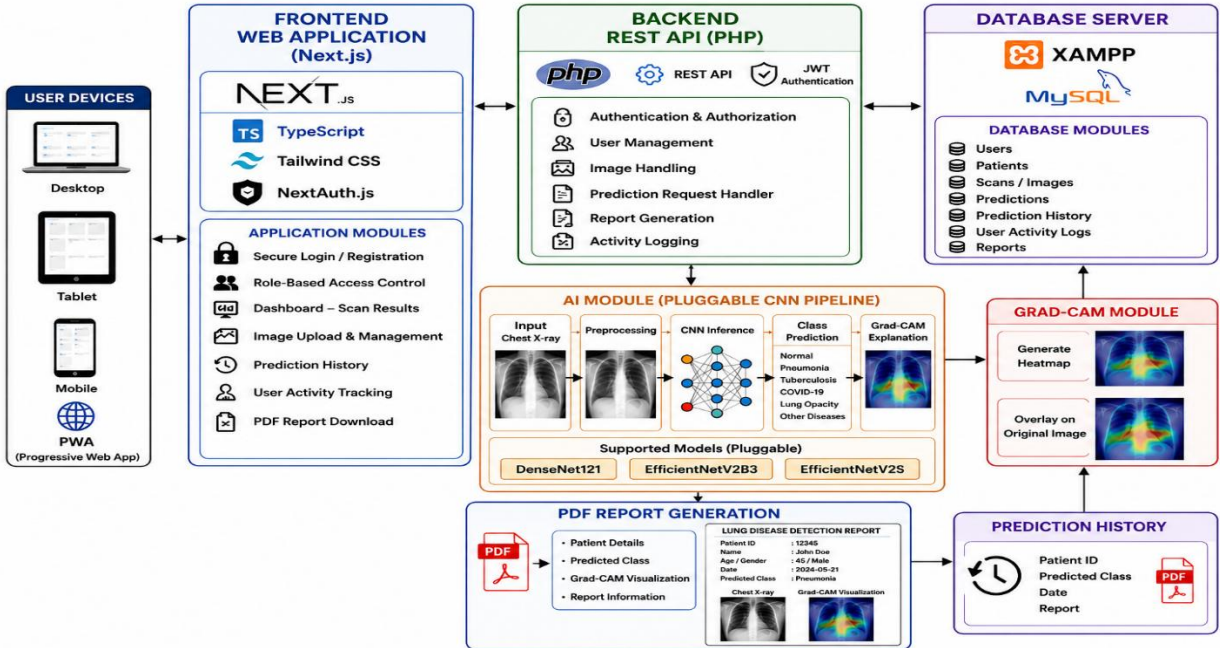


Fig. 1. End-to-End System Deployment Architecture of the Web-Based Lung Disease Detection Platform.

The system is deployed as a web-based, mobile-friendly lung disease detection platform for desktops, tablets, and mobile devices. The frontend uses Next.js, TypeScript, Tailwind CSS, and NextAuth, while the backend uses PHP REST APIs, JWT authentication, and MySQL via XAMPP, providing secure login, role-based access control (patient, doctor, researcher, administrator), an administrative panel for managing users, scans, activity, and statistics, and account self-management (profile editing, password change and recovery, and deletion). Dashboards present scan results, confidence visualization, image management, prediction history, and CSV export. Inference is decoupled into a dedicated Python (FastAPI) microservice that loads the trained Keras models and runs a pluggable CNN pipeline supporting DenseNet121, EfficientNetV2B3, or EfficientNetV2S, performing preprocessing, prediction, and confidence scoring. A post-hoc out-of-distribution safeguard, combining Mahalanobis-distance, energy, and maximum-softmax signals, withholds the diagnosis for invalid inputs, recording the verdict, score, and reason. Results are stored, and automated PDF reports in English, Sinhala, and Tamil contain patient details, predicted class, confidence, and timestamp. An assistance chatbot, light and dark modes, and informational pages complete a secure end-to-end clinical decision support system. Fig. 1 illustrates the end-to-end system deployment architecture of the proposed web-based lung disease detection platform, encompassing the AI prediction pipeline, out-of-distribution safeguard, secure user authentication, and automated multilingual PDF report generation.

IV. RESULTS AND COMPARATIVE ANALYSIS

The comparative evaluation of DenseNet121, EfficientNetV2B3, and EfficientNetV2S demonstrates competitive performance across all three architectures in the four-class chest X-ray classification task. DenseNet121 achieved the highest accuracy of 85.31%, indicating that dense feature reuse and strong gradient propagation are effective for identifying chest disease patterns. EfficientNetV2B3 followed closely with 84.30%, with a performance gap of only 1.01% attributed to its balanced scaling of depth, width, and resolution enabling efficient capture of both fine-grained and global radiographic features, confirming it as a strong candidate for clinical deployment where accuracy and computational efficiency are both required. EfficientNetV2S achieved 80.54%, the lowest among the three models, yet demonstrates that Fused-MBConv-based architectures remain viable for multiclass chest X-ray classification and remain practically useful in lightweight deployment settings where computational efficiency is prioritized. In a real-world clinical deployment, accuracy is not the sole criterion. Inference speed, memory requirements, scalability, and compatibility with web or mobile platforms are also important considerations especially for real-time diagnosis, patient safety, and system efficiency evaluation.

TABLE II. COMPARISON OF DENSENET121, EFFICIENTNETV2B3, AND EFFICIENTNETV2S FOR FOUR-CLASS CXR CLASSIFICATION USING THE AUTHORS' TRAINED MODELS, SHOWING OVERALL ACCURACY ON THE CURATED DATASET.

Model	Accuracy (%)	Precision	Recall	F1-Score
DenseNet121	85.31	0.84	0.85	0.84
EfficientNetV2B3	84.30	0.83	0.83	0.83
EfficientNetV2S	80.54	0.80	0.82	0.80

Table II presents the comparative performance of DenseNet121, EfficientNetV2B3, and EfficientNetV2S in terms of accuracy, precision, recall, and F1-score, demonstrating that DenseNet121 achieved the highest overall accuracy of 85.31%, followed closely by EfficientNetV2B3 at 84.30% and EfficientNetV2S at 80.54%. EfficientNetV2S therefore remains a practical candidate for lightweight deployment scenarios even if its accuracy is marginally lower than the other two models. The results collectively support the feasibility of the proposed four-class classification approach. Unlike binary models that only separate normal from abnormal cases, this framework provides more specific diagnostic categories. Such classification is clinically meaningful because bacterial pneumonia, viral pneumonia, and COVID-19 may require different treatment decisions, isolation protocols, and follow-up procedures. Although final diagnosis must remain under medical supervision, automated multi-class classification can meaningfully assist in triage and preliminary screening.

TABLE III. PROPOSED FUNCTIONAL COMPONENTS OF THE CLINICAL DECISION SUPPORT SYSTEM

Component	Function
Image Upload	Accepts chest X-ray images through the web interface
CNN Prediction Module	Classifies images into Normal, COVID-19, Bacterial Pneumonia, or Viral Pneumonia
Confidence Visualization	Displays class probability and prediction confidence
Grad-CAM Explanation	Highlights image regions influencing the prediction
Prediction History	Stores previous scan results for patient management
PDF Report Generator	Produces structured medical reports
Web Dashboard	Supports user, scan, and result management
Mobile-Friendly PWA	Enables app-like access across devices
Secure Authentication	Uses JWT authentication and role-based access control

Table III presents the main functional components of the proposed clinical decision support system and their respective functions. The integration of Grad-CAM improves interpretability by allowing users to inspect whether the model focuses on relevant lung regions during prediction. This is particularly important when two models achieve similar accuracy levels. A model with slightly lower accuracy but more clinically meaningful heatmaps may be more useful than a model with marginally higher accuracy but unreliable or diffuse attention patterns. Final model selection should therefore consider both quantitative performance metrics and the qualitative clinical relevance of Grad-CAM explanations.

V. DISCUSSION

The findings demonstrate that explainable transfer learning provides a practical approach for multiclass pneumonia detection from chest X-ray images. DenseNet121 achieved the best overall accuracy, though the small performance gap among architectures indicates that dataset quality, preprocessing, augmentation, class balance, and fine-tuning configuration are equally critical alongside architecture selection. DenseNet121's dense connectivity enables multi-level feature reuse advantageous for detecting subtle chest X-ray disease indicators. EfficientNetV2B3 closely followed with 84.30% accuracy, confirming compound-scaled architectures suit deployment-oriented medical imaging, while EfficientNetV2S at 80.54% may offer advantages in lightweight deployment scenarios pending further validation across diverse datasets. Beyond classification, the proposed

system integrates clinically relevant features including web and mobile-friendly deployment, JWT authentication, role-based access control, historical patient management, and automated PDF report generation. Grad-CAM improves transparency by verifying model attention on relevant lung regions, though it should be treated as supportive visual evidence rather than definitive diagnostic explanation. The framework has practical potential in screening and triage environments; however, external validation across different hospitals, devices, and demographic groups is essential before clinical deployment. The system is therefore appropriately positioned as a decision-assistance tool that supports rather than replaces medical judgment.

VI. CONCLUSION

This paper presented an explainable deep learning framework for pneumonia detection and classification using chest X-ray images. The study addressed a four-class classification problem involving Normal, COVID-19, Bacterial Pneumonia, and Viral Pneumonia cases. Three convolutional architectures DenseNet121, EfficientNetV2B3, and EfficientNetV2S were evaluated under a unified transfer learning framework. DenseNet121 achieved the highest accuracy of 85.31%, followed by EfficientNetV2B3 with 84.30% and EfficientNetV2S with 80.54%. Beyond model comparison, the proposed work contributes an end-to-end clinical decision support framework that integrates prediction, Grad-CAM explainability, historical prediction management, mobile-friendly access, secure authentication, and automated PDF report generation. This combination addresses important gaps in existing studies, which often focus only on classification accuracy without considering interpretability and practical deployment requirements.

The results indicate that explainable deep learning can meaningfully support chest X-ray-based pneumonia classification, but clinical use requires careful external validation, expert oversight, and responsible deployment practice. The proposed system is best viewed as an assistive tool that supports clinicians by providing rapid predictions and interpretable visual evidence while clinical authority remains firmly with medical professionals. This study has several limitations. The evaluation was conducted using a single chest X-ray dataset, which may limit the generalizability of results to external datasets with different acquisition characteristics. Furthermore, although Grad-CAM improves model interpretability, its visual explanations may not always precisely correspond to the clinically relevant regions that an experienced radiologist would prioritize. Future work will focus on external validation using diverse chest X-ray datasets, integration of multimodal clinical data to improve diagnostic accuracy, exploration of advanced explainability methods beyond Grad-CAM, and model optimization techniques for practical healthcare deployment in resource-constrained environments.

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